



Full Length Article

Bandgap mechanism and seismic wave attenuation performance of buried seismic metamaterials with double resonators

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ABSTRACT

The utilization of the band gap properties in periodic structures can control the propagation of elastic waves, leading to a focus in recent years on seismic metamaterials, but it is challenging to obtain ultrawide low-frequency bandgaps. To broaden the low-frequency band gaps, two approximately equivalent buried seismic metamaterials with double resonators (BMDR) were proposed based on theoretical derivations and analyses. The mechanisms behind band gap formation and seismic wave attenuation capabilities were thoroughly investigated and validated through finite element calculations. The results show that the BMDR can generate a low-frequency wide band gap and exhibit strong attenuation of seismic waves within the band gap range. On average, the BMDR attenuated seismic waves by approximately 80 % within the local resonance band gap and by about 40 % within the Bragg band gap without amplifying seismic responses outside these band gaps.

1. Introduction

The utilization of periodic structures based on Bragg scattering phenomena [1–3] and localized resonance mechanisms [3–6] to deploy wave barriers such as piles, trenches, and continuous walls near buildings is an effective method to block the propagation of elastic waves, which can be used to mitigate environmental vibrations or seismic ground motions. The previous studies [7–11] have demonstrated that mechanical metamaterials containing multiple resonators can exhibit multiple band gaps, in contrast to metamaterials with a single resonator.

Numerous studies have been conducted on periodic metamaterials with a single resonator, characterized by diverse structural forms and varying filtering efficacy [12–16]. Xu et al. [17] compared two structural forms of metamaterials, namely hollow cylinders and hollow cylinders with added cross-shaped elements. They found that the incorporation of cross-shaped structures increased the band gap width by approximately threefold, even when layered soil was taken into account. Ni et al. [18] designed a periodic pile barrier and comprehensively discussed the key influencing parameters. Subsequently, the correction of the bandgap for the periodic pile barriers was validated through frequency and time domain analysis. However, the theoretical derivation of the dispersion relation was not provided or was insufficient, limiting their study to numerical simulations. Zhao et al. [19] studied four periodic pile barrier structures with the same fill ratio but different cross-sections based on the Bloch-Floquet theory. In the periodic potential barrier composed of cross-sectional foam plastic, surface wave band gaps were observed within the 9.64 to 13.92 Hz range. Lou et al. [20]

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designed and studied a gradient amplitude-modulated rod to achieve ultra-wideband vibration attenuation and rainbow reflection. The study highlights the importance of incorporating a specific number of identical units in each section and introducing mild damping in each resonator to effectively suppress ultra-wideband vibrations.

Mu et al. [21] proposed a two-dimensional periodic foundation system utilizing steel-silicon nested components as local resonators. In addition, the correctness of the band gap range obtained through theoretical derivation was demonstrated by experiments. Du et al. [22] designed a seismic metamaterial using an H-shaped fractal column and conducted numerical simulations to analyze the propagation of seismic surface waves within a structured band. It was revealed that the new energy levels associated with the fractal structures of seismic metamaterials expanded the overall band gap. Although structural transformations and changes in materials can widen the band gap of metamaterials and improve the filtering effectiveness, the degree of improvement is limited.

The double resonator metamaterial is based on acoustic metamaterials' band gap coupling characteristics. It combines two heterogeneous unit cells of seismic metamaterials arranged periodically. The low-frequency band gaps are merged and broadened through the coupling resonance of these heterogeneous unit cells, thereby enhancing the attenuation of low-frequency seismic waves. In recent years, research has been increasing on dual-resonator periodic metamaterials to overcome the bottleneck of narrow bandgaps in metamaterials with single-resonator [23–26]. Gao et al. [27] theoretically derived the dispersion curves of dual-resonator periodic structures, revealing that coupling Bragg scattering and local resonance could achieve a wider bandgap. Zhu et al. [28] proposed an Electromagnetic Metamaterial EMM beam with a band cross-sectional distribution resonator, widening the bandgap and reducing vibration. Zhou et al. [29] studied a continuous medium model of a two-dimensional metamaterial lattice, establishing three continuous medium models. It was found that the dispersion relationship and bandgap characteristics of metamaterials could be more accurately represented by appropriately selecting non-local parameters and employing non-local gradient continuum models. Ghanvloo et al. [30] derived the dispersion relationship of elastic waves in one-dimensional dual-resonator structures and obtained a continuum model using differential-difference equations. They also demonstrated the model's applicability for predicting the dispersion relationship of one-dimensional dual-resonator structures. Alamri et al. [31] proposed an acoustical metamaterial with double resonators and a theoretical investigation was conducted on the effect of damping on asymmetric wave transmission, numerically and experimentally. Experimental results showed that the combined effect of dissipation damping significantly increased the bandgap width in the asymmetric transmission region. Jiang et al. [32] proposed a multi-resonator periodic structure with local resonance characteristics. The low-frequency bandgap structures were analyzed and compared using the finite element and circuit analogy methods. It was found that the multi-resonator periodic structure could couple the bandgaps generated by a single resonator, and the influence of structural factors on the bandgap was analyzed. The structure was divided into an average of 200 parts, exhibiting a wider effective bandgap below 500 Hz, and reducing the spacing between unit structures could increase the strength of structural coupling. Kim et al. [33] proposed a dual-resonance sound absorption structure with porous media, three samples were fabricated and compared for analysis. This paper combines the above studies to explore the bandgap mechanism of dual-resonator metamaterials and their application in attenuating seismic waves.

The research on periodic structures with dual-resonators mentioned above primarily focused on investigating the effects of parameters such as resonator mass, spring stiffness, and damping on the bandgap, without further exploring their potential applications. Although research on three-dimensional seismic metamaterials has been extensive, two-dimensional approaches offer a more direct means of simulating and visualizing wave propagation patterns within these materials. Before extending complexities to three dimensions, such simplification is crucial for isolating and understanding the fundamental mechanisms. Therefore, this study introduces two types of buried metamaterials with dual resonance cavities from a two-dimensional perspective: buried metamaterial with double resonators-1(BMDR-1) and buried metamaterial with double resonators-2(BMDR-2). BMDR-1 is constructed using theoretically derived parameters, where springs are replaced with trusses and mass units with steel materials, resulting in a double-resonator metamaterial with distinct unit cells. BMDR-2 is the physical equivalent of BMDR-1, utilizing rubber instead of trusses. Both structures produce low-frequency band gaps and attenuation domains corresponding to theoretical analysis, demonstrating effective attenuation of seismic waves. Compared to previous studies, the two metamaterial architectures feature well-defined geometric and material parameters, coupled with a streamlined design that eliminates redundant structural branches while strategically retaining components critical for seismic wave attenuation. The inner and outer oscillators are interconnected via four-directional trusses or

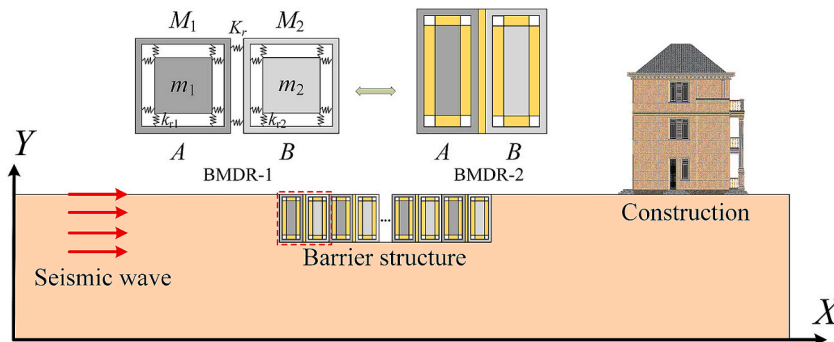


Fig. 1. Metamaterial model.

rubber connectors, which substantially expand the low-frequency locally resonant bandgaps (0.5–10 Hz). This optimization provides a novel design paradigm for practical applications in seismic metamaterials, particularly for earthquake-resistant infrastructure.

The organizational structure of this paper is as follows. Section 2 introduces the models of the two buried dual-resonator metamaterials and provides their equivalent theoretical models. Section 3 conducts dispersion analysis on the two BMDRs and compares the dispersion curves with the theoretical derivation results. Finally, the attenuation effects of the BMDRs on seismic waves across various dominant frequencies are obtained through time-domain analysis.

2. Theoretical formulas

As shown in Fig. 1, excitation propagates along the X-axis at the near free surface, with subterranean metamaterial arranged along the X-axis underground. The metamaterial is assumed to extend infinitely along the Z-axis direction for simplifying analytical calculations, while the soil is modeled as an isotropic and homogeneous linear elastic material. This paper presents two metamaterial models. In BMDR-1, the resonant unit cell consists of internal resonators with masses m_1 and m_2 and external resonators with masses M_1 and M_2 , interconnected horizontally and vertically by springs with stiffness k_{r1} and k_{r2} , respectively. The two external resonators are connected by a spring with stiffness K_r . By adjusting parameters such as structure density, dimensions, Young's modulus, and Poisson's ratio, the structure can be equivalently represented as the solid model BMDR-2, where the external and internal masses are composed of steel and interconnected by rubber. The material properties of BMDR-1 and BMDR-2 correspond numerically to each other. BMDR-1 represents a finite element model that more closely aligns with theoretical models, and its construction using actual materials is similar to that of BMDR-2.

To derive the dispersion equation of the BMDR, the theoretical equivalent model in the X-direction is illustrated in Fig. 2. Within the n th resonant unit, the displacements of the two external resonators are denoted as S_n and W_n . In contrast, those of the two internal resonators are represented as s_n and w_n . The external resonators are interconnected by a spring with stiffness K , whereas the external resonators are connected to the internal resonators via springs with stiffness k_1 and k_2 . The lattice constant of the unit cell is denoted as d .

For the metamaterial structure with an infinite array, the motion equation of the n th resonant unit cell is expressed as follows

$$\begin{cases} M_1 \frac{d^2 S_n}{dt^2} + 2KS_n - KW_n - KW_{n-1} + 2k_1 S_n - 2k_1 s_n = 0 \\ M_2 \frac{d^2 W_n}{dt^2} + 2KW_n - KS_n - KS_{n+1} + 2k_2 W_n - 2k_2 w_n = 0 \\ m_1 \frac{d^2 s_n}{dt^2} + 2k_1 S_n - 2k_1 s_n = 0 \\ m_2 \frac{d^2 w_n}{dt^2} + 2k_2 W_n - 2k_2 w_n = 0 \end{cases} \quad (1)$$

It is assumed that the harmonic solution form of the motion equation is expressed as follows:

$$\begin{cases} S_n = S e^{i(nkd - \omega t)} \\ W_n = W e^{i(nkd - \omega t)} \\ s_n = s e^{i(nkd - \omega t)} \\ w_n = w e^{i(nkd - \omega t)} \end{cases} \quad (2)$$

In the equation, k represents the wave number, i , ω , and t are the imaginary unit, the angular frequency, and time. Substituting Eq. (2) into Eq. (1), we obtain

$$\begin{bmatrix} -M_1 \omega^2 + 2K + 2k_1 & -2k_1 & -K(1 + e^{-ikd}) & 0 \\ -2k_1 & -m_1 \omega^2 + 2k_1 & 0 & 0 \\ -K(1 + e^{ikd}) & 0 & -M_2 \omega^2 + 2K + 2k_2 & -2k_2 \\ 0 & 0 & -2k_2 & -m_2 \omega^2 + 2k_2 \end{bmatrix} \begin{bmatrix} S \\ s \\ W \\ w \end{bmatrix} = 0 \quad (3)$$

If the equation has a solution, then its coefficient determinant is equal to 0, that is:

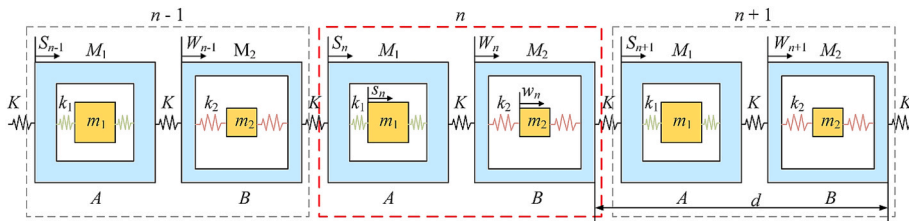


Fig. 2. Theoretical model.

$$\det \begin{bmatrix} -M_1\omega^2 + 2K + 2k_1 & -2k_1 & -K(1 + e^{-ikd}) & 0 \\ -2k_1 & -m_1\omega^2 + 2k_1 & 0 & 0 \\ -K(1 + e^{ikd}) & 0 & -M_2\omega^2 + 2K + 2k_2 & -2k_2 \\ 0 & 0 & -2k_2 & -m_2\omega^2 + 2k_2 \end{bmatrix} = 0 \quad (4)$$

Solving the equation gives:

$$\begin{aligned} & \left[\left(\frac{2k_1m_1 + 2k_1M_1 - m_1M_1\omega^2}{2k_1M_1 - m_1M_1\omega^2} \right) \left(\frac{M_2}{M_1} + \frac{2k_1m_2}{2k_2M_1 - m_2M_1\omega^2} \right) \right] \left(\frac{M_1\omega^2}{K} \right)^2 \\ & - 2 \left[\left(\frac{2k_1m_1 + 2k_1M_1 - m_1M_1\omega^2}{2k_1M_1 - m_1M_1\omega^2} \right) + \left(\frac{M_2}{M_1} + \frac{2k_1m_2}{2k_2M_1 - m_2M_1\omega^2} \right) \right] \left(\frac{M_1\omega^2}{K} \right) - e^{ikd} - e^{-ikd} + 6 = 0 \end{aligned} \quad (5)$$

Let $G_n = \frac{2k_n m_n}{2k_n M_n - m_n M_1 \omega^2}$, Eq. (5) can be simplified to

$$\text{Im}(kd) = \text{arcsec} \left[\frac{2}{\left(\frac{M_1\omega^2}{K} (1 + G_1) - 2 \right) \left(\frac{M_1\omega^2}{K} \left(\frac{M_2}{M_1} + \frac{k_1}{k_2} G_2 \right) - 2 \right) + 2} \right] \quad (6)$$

Eq. (6) represents the dispersion equation of the resonant unit. The bandgap's starting frequency corresponds to the structure's resonant frequency. The masses M_1 , m_1 , M_2 , and m_2 of the n th resonant unit are represented as effective masses, as shown in Fig. 3.

From Eq. (2), it can be derived that

$$\begin{cases} \frac{s}{S} = \frac{2k_1}{2k_1 - \omega^2 m_1} \\ \frac{w}{W} = \frac{2k_2}{2k_2 - \omega^2 m_2} \end{cases} \quad (7)$$

The effective mass, denoted as $M_{1,eff}$ or $M_{2,eff}$, can be defined as a mass model that possesses the same momentum as the equivalent pre-resonant unit, namely:

$$\begin{cases} M_{1,eff} \frac{dS}{dt} = M_1 \frac{dS}{dt} + m_1 \frac{ds_1}{dt} \\ M_{2,eff} \frac{dW}{dt} = M_2 \frac{dW}{dt} + m_2 \frac{dw_1}{dt} \end{cases} \quad (8)$$

Substituting Eq. (7) into Eq. (8), we obtain

$$\begin{cases} M_{1,eff} = M_1 + \frac{2k_1 m_1}{2k_1 - m_1 \omega^2} \\ M_{2,eff} = M_2 + \frac{2k_2 m_2}{2k_2 - m_2 \omega^2} \end{cases} \quad (9)$$

The mass matrix of the n th resonant unit is:

$$[M] = \begin{bmatrix} M_{1,eff} & 0 \\ 0 & M_{2,eff} \end{bmatrix} \quad (10)$$

The stiffness matrix is:

$$[K] = \begin{bmatrix} 2K & -2K \\ -2K & 2K \end{bmatrix} \quad (11)$$

The displacement vector is

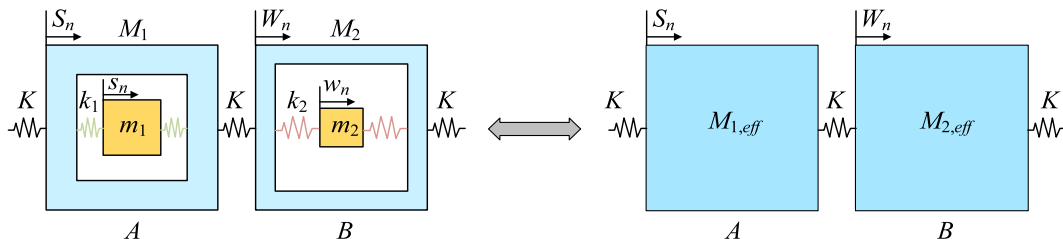


Fig. 3. Equivalent model.

$$\{x(t)\} = \begin{Bmatrix} A_1 \\ A_2 \end{Bmatrix} \sin(\omega t + \theta) \quad (12)$$

Where A_1 and A_2 are the displacement amplitudes of the two equivalent masses, the equation of motion for the resonant unit is obtained as follows.

$$[M]\{\ddot{x}(t)\} + [K]\{x(t)\} = 0 \quad (13)$$

The eigen equation is expressed as:

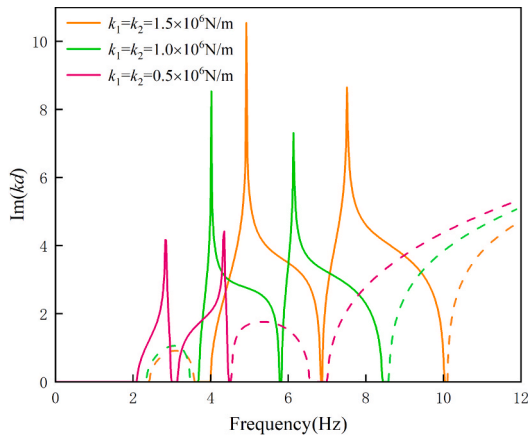
$$|[K] - \omega^2[M]| = 0 \quad (14)$$

Both:

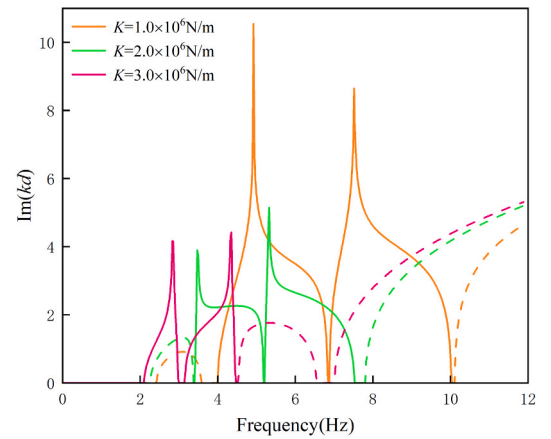
$$\omega^2 M_{1,eff} M_{2,eff} = 2K(M_{1,eff} + M_{2,eff}) \quad (15)$$

From the eigen equation, we can obtain the natural frequency inside the resonators, which is also the starting frequency of the local resonance bandgap.

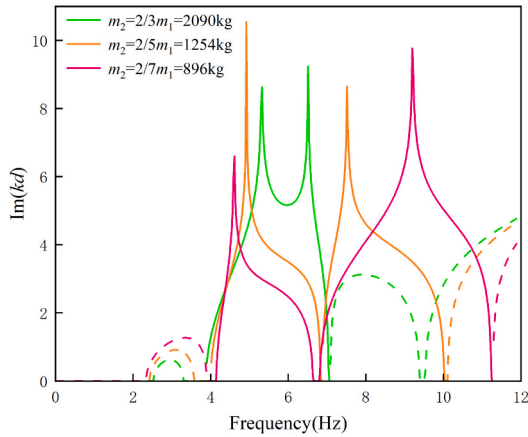
$$\omega_{1,2,on} = \sqrt{\frac{2K(M_{1,eff} + M_{2,eff})}{M_{1,eff} M_{2,eff}}} \quad (16)$$



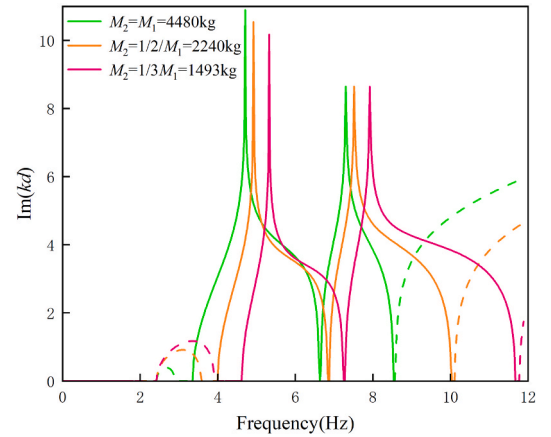
(a)



(b)



(c)



(d)

Fig. 4. Parameter analysis (a) The impact of k_1 and k_2 on the dispersion curve; (b) The impact of K on the dispersion curve; (c) The impact of m_1 and m_2 on the dispersion curve; (d) The impact of M_1 and M_2 on the dispersion curve.

The cutoff frequency of the bandgap corresponds to the frequency at which the curve intersects the axis. Substituting $k = 0$ into Eq. (6), we obtain:

$$\left[\frac{M_1 \omega^2}{K} (1 + G_1) - 2 \right] \left[\frac{M_1 \omega^2}{K} \left(\frac{M_2}{M_1} + \frac{k_1}{k_2} G_2 \right) - 2 \right] = 0 \quad (17)$$

Solved

$$\omega_{1, \text{off}} = \frac{k_1 M_1^2 + m_1 M_1 (k_1 + K) \pm \sqrt{k_1^2 M_1^4 + 2k_1 m_1 M_1^3 (k_1 - K) + m_1^2 M_1^2 (k_1 + K)^2}}{m_1 M_1^2} \quad (18)$$

$$\omega_{2, \text{off}} = \frac{k_2 M_2^2 + m_2 M_1 (k_1 + K) \pm \sqrt{k_2^2 M_2^4 + 2k_2 m_2 M_1 M_2^2 (k_1 - K) + m_2^2 M_1^2 (k_1 + K)^2}}{m_2 M_1 M_2}$$

According to Eq. (16), as K decreases, both the starting frequencies $\omega_{1, \text{on}}$ and $\omega_{2, \text{on}}$ generated by localized resonances decrease, causing the bandgap to shift towards lower frequencies.

According to Eq. (6), the variation in band gap ranges is primarily determined by the structure's equivalent mass M_E and equivalent stiffness K_E . This study analyzes the effects of parameters m_1 , m_2 , k_1 , k_2 , M_1 , M_2 , and K on the imaginary part of the dispersion curves, with results presented in Fig. 4.

Local resonance band gaps are generated by the local resonance of resonators. For a resonator to resonate, the excitation frequency must match its natural frequency, which is determined by its effective mass and stiffness. The starting frequency of the Bragg band gap depends solely on the periodicity of the system. In contrast, the cutoff frequency of the band gap is influenced by both the system's periodicity and the resonator parameters [34]. According to Bragg's law, strong reflection occurs when the wavelength λ and period d satisfy the Bragg condition $2d \sin(\theta) = m\lambda$, where m is the Bragg order, typically $m = 1$. For optical or acoustic waves, the relationship between frequency f and wavelength λ is given by $f = \frac{c}{\lambda}$, where c is the wave propagation speed in the medium. Consequently, a larger period corresponds to a longer Bragg reflection wavelength (i.e., λ increases), resulting in a lower frequency, which in turn lowers the starting and cutoff frequencies of the bandgap.

In the dispersion curve, the band gap generated by local resonance exhibits sharp maximum attenuation at a certain frequency, and the attenuation curve within the band gap is asymmetrical. This characteristic of resonance band gaps has been commonly observed in previous studies on local resonance structures [35,36]. A structure with several resonant cavities typically will have multiple local resonance band gaps, provided there is no coupling [25]. However, for Bragg band gaps, the curve always changes smoothly within the band gap region [34,37].

The solid lines in the Fig. 4 represent band gaps induced by the local resonance of resonant units, while the dashed lines represent Bragg band gaps. As shown in Fig. 4(a), although the starting frequencies of the two local resonance band gaps shift to higher frequencies with the increase of k_1 and k_2 , the band gap range and attenuation effect also significantly increase. Conversely, as K increases, it suppresses the vibration of the structure, resulting in a gradual decrease in both the band gap range and attenuation effect, as illustrated in Fig. 4(b). Assuming values for m_1 and M_1 are 3136 kg and 4480 kg, respectively, we then discuss the variation in the band gap range with changes in m_2 and M_2 . Fig. 4(c) shows that as m_2 decreases, the band gap range gradually shifts to higher frequencies. However, when m_2 approaches m_1 too closely, the two local resonance band gaps couple together, reducing the overall band gap range. As M_2 decreases, the effective mass of the structure gradually decreases, leading to an increase in the starting frequency of the band gap, thus shifting the band gap range to higher frequencies, as illustrated in Fig. 4(d). The parameter values derived from the above analysis are listed in Table 1 [38,39].

3. Finite element analysis of BMDRs

In this section, finite element models were established using finite element software to verify the correctness of theoretical derivations and the filtering characteristics after forming the barrier with periodically arranged unit cells. Dispersion, frequency domain, and time history analyses were performed on these models.

Table 1
Material parameter.

m_1 /(kg)	m_2 /(kg)	M_1 /(kg)	M_2 /(kg)	K /(N/m)
3136	1254	4480	2240	1.0×10^6
E_0 /(GPa)	ν_0	K_r /(N/m)	k_{1r} /(N/m)	k_{2r} /(N/m)
209	0.275	0.5×10^6	0.75×10^6	0.75×10^6
ρ_2 /(kg/m ³)	ν_2	E /(MPa)	ρ /(kg/m ³)	ν
1300	0.4	46	1800	0.25

3.1. Dispersion analysis of BMDRs

Fig. 5(a) shows a single-unit cell model of the buried metamaterial with double resonators (BMDRs). The lattice constant of the single cell in the metamaterial is denoted as d . In this model, both BMDR-1 and BMDR-2 consist of external and internal mass blocks made of steel, with masses designated as M_1, M_2, m_1 , and m_2 respectively, consistent with the theoretical analysis presented earlier. The elastic modulus of the materials is denoted as E_0 , and the Poisson's ratio is denoted as ν_0 . BMDR-1 is modeled using truss elements with stiffness values set as $K_r = 0.5 K$, $k_{1r} = k_{2r} = 0.5 k_1 = 0.5 k_2$. Furthermore, BMDR-2 is simulated using a rubber structure instead of truss elements, with a density denoted as ρ_2 and a Poisson's ratio of ν_2 . The axial elastic modulus is determined using the formula $K = EA/L$, ensuring equivalence with the stiffness of each truss in BMDR-1, with the material parameters specified in Table 1. A local enlargement of the solid model of BMDR-2 is illustrated in Fig. 5(b), with geometric parameters provided in Table 2. The elastic modulus, density and Poisson's ratio of the soil are denoted as E, ρ , and ν . The resonant frequency of the structure is set as $f_r = 4$ Hz. By considering the given parameters of the soil and combining them, the Rayleigh wavelength of the resonant substructure is determined as $\lambda_r = 23$ m, satisfying the conditions for the application of the effective medium method ($\lambda_r \gg d$) [40,41].

Figs. 6(a) and 6(b) show the theoretical and numerical band gaps of the BMDR. The numerical band gaps of BMDR-1 and BMDR-2 are in reasonable agreement with the primary band gaps derived from theoretical analysis, validating that the theoretical band gap solution can, to some extent, approximately predict the range of numerical band gaps. BMDR-1 displayed three distinct band gaps occurring at frequency ranges of 2.06–3.17 Hz, 4.04–6.43 Hz, and 6.89–9.60 Hz, among which the band gap at 2.06–3.17 Hz stemmed from the Bragg scattering mechanism, while those at 4.04–6.43 Hz and 6.89–9.60 Hz were attributed to the local resonance mechanism. Similarly, BMDR-2 has three band gaps spanning from 2.07–3.47 Hz, 4.03–6.30 Hz, to 7.17–9.50 Hz, where the band gap at 2.07–3.47 Hz originated from the Bragg mechanism, whereas those at 4.03–6.30 Hz and 7.17–9.50 Hz emerged due to the local resonance mechanism. From the error range of the bandgap in Fig. 6(c), it can be observed that the bandgap ranges for BMDR-1 and BMDR-2 are roughly the same as the theoretical bandgap range, and the initial frequency prediction of the second and third band gaps is more accurate. Specifically, the first bandgap width is approximately the same, but the center frequency of the bandgap decreased by 0.38 Hz and 0.22 Hz compared to the theoretical bandgap. The second and third bandgap widths have slightly reduced, with error rates of 14.9 % and 14.5 % for BMDR-1, and 19.2 % and 26.2 % for BMDR-2. All error rates fall within the acceptable range for this study. The bandgap results reveal that the center frequencies of the bandgaps for BMDR-1 and BMDR-2 are significantly lower than those reported in previous studies[39,42], indicating superior performance in the lower frequency range. Additionally, the number of bandgaps for both structures has noticeably increased, enabling them to cover a broader frequency range, which enhances their adaptability in multi-band applications. These characteristics make BMDR-1 and BMDR-2 more flexible and practical for real-world applications, particularly in scenarios that require both wide bandwidth and low-frequency capabilities, where they demonstrate a clear advantage.

The mechanisms underlying the band gap generation are closely related to the excitation frequency f and the structural frequency. The vibration modes corresponding to the frequencies in the soil are excited as the excitation frequency f gradually increases,

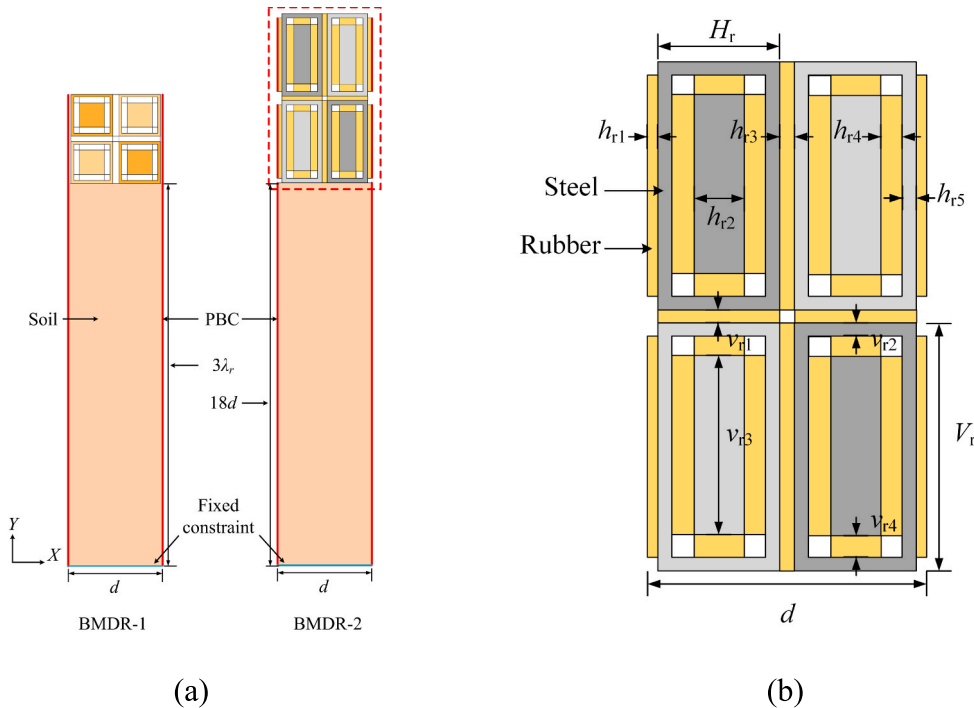


Fig. 5. Unit cell model (a) Single unit cell model of BMDR; (b) Partial magnification of BMDR-2.

Table 2
Geometric parameter.

$H_r/(m)$	$h_{r1}/(m)$	$h_{r2}/(m)$	$h_{r3}/(m)$	$h_{r4}/(m)$	$h_{r5}/(m)$
1	0.05	0.4	0.1	0.2	0.1
$V_r/(m)$	$v_{r1}/(m)$	$v_{r2}/(m)$	$v_{r3}/(m)$	$v_{r4}/(m)$	$d/(m)$
2	0.1	0.1	1.4	0.2	2.2

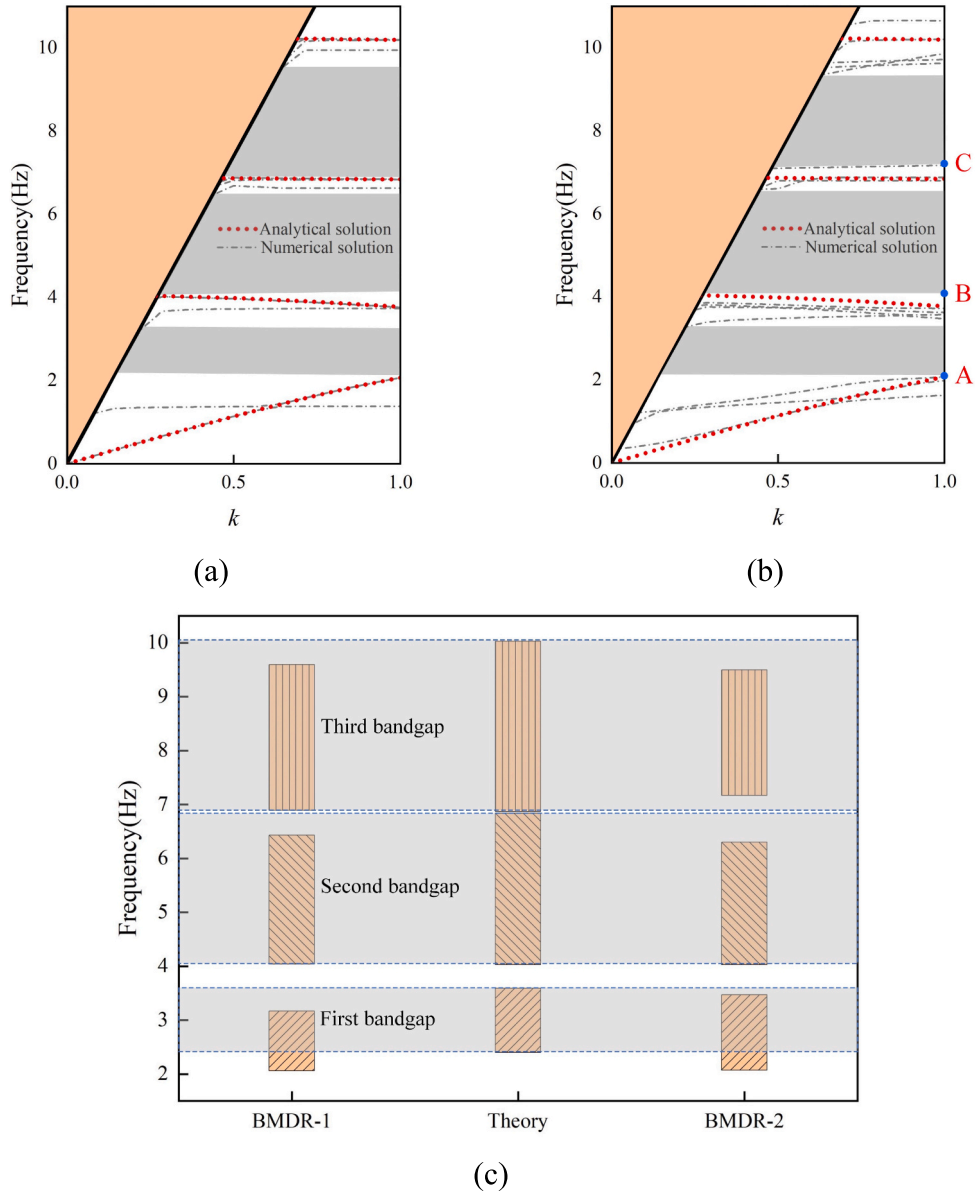


Fig. 6. Dispersion curve (a) BMDR-1; (b) BMDR-2; (c) Bandgap comparison.

becoming the dominant mode of vibration, and consequently, no band gaps occur.

However, as the excitation frequency f approaches the resonant frequency of the resonator, the corresponding mode gradually becomes the principal mode of the resonator. If the resonator mode closely matches the mode of the soil, a strong coupling effect occurs, which constrains soil vibration propagation and leads to the formation of band gaps. In such modes, the resonators' equivalent mass and stiffness determine the initial frequency of the band gap.

$$f = \frac{1}{2\pi} \sqrt{K_E/M_E} \quad (19)$$

Where, K_E and M_E represent the equivalent stiffness and mass of the resonator for this mode, respectively.

As the excitation frequency f increases, the frequency deviates gradually from the resonator frequency within this mode, reducing the restrictive effect on the soil mode. Once the frequency completely diverges from that of the resonator in this mode, the resonator no longer restricts the soil mode, thus marking the cut-off frequency.

Points were selected from the theoretical solution curve, and the corresponding vibration modes of the structure in the numerical solution were analyzed to elucidate the vibrational characteristics of the structure during local resonance and Bragg scattering. Extensive analysis in this study has shown that when the selected points lie within the local resonance band gap, a sharply varying curve is observed, and the structure typically experiences significant vibrations in certain regions, distinct from other areas. When the points are within the Bragg band gap, where the curve exhibits smooth variation, the structure does not induce strong local resonance in any specific region, the overall vibrational behavior of the structure is influenced by collective interactions within the periodic arrangement. The vibration modes of the metamaterial structure at points A, B, and C, as shown in Fig. 6(b), are used to explain the mechanism behind the band gap formation in BMDR, as illustrated in Fig. 7. At point B, vibrations primarily occur at the positions of two internal oscillators with lower natural frequencies within the structure, inducing horizontal resonance to dissipate energy, consequently opening up the band gap. At point C, the external and internal vibrators vibrate in opposite directions to dissipate energy. Point A does not exhibit localized resonance but instead generates the band gap through Bragg scattering, which results in weaker attenuation effects on wave energy.

3.2. Attenuation performance of BMDRs

Earthquakes transfer energy in the form of seismic waves, which can be categorized into body waves and surface waves. Body waves reach the Earth ground surface and then generate Rayleigh waves, which are characterized by large amplitudes, slow attenuation, and long propagation distances. These waves are the primary contributors to structural damage during an earthquake. This paper primarily focuses on the attenuation of Rayleigh waves by metamaterial structures.

Figs. 8 (a) and 8(b) show the two-dimensional models of BMDR-1 and BMDR-2 which are periodically arranged in the X and Y directions and extend indefinitely along the vertical XoY plane. The burial depth of the metamaterial is λ_r and $2d$, with a width of $2\lambda_r$ and $10d$. To separate body and surface waves, the distance from the resonant unit cell to the excitation point is set as 4λ and $20d$, ensuring that only Rayleigh waves affect the metamaterial barrier. In addition, the other parameters such as excitation points, output points, and source distances are indicated in the figure. Output points are set on a concrete frame structure to demonstrate the attenuation of seismic waves by the metamaterial by calculating the acceleration response at different heights.

A unit harmonic excitation is applied at the excitation points to perform frequency domain analysis on the buried metamaterial with double resonators. The attenuation effect of the metamaterial on excitations at different frequency bands is expressed using the acceleration transfer spectrum. To prevent the reflection of elastic waves at the boundaries, perfect matching layers with a thickness of λ_r and $3d$ are applied to the outer side of the soil model.

Simultaneously, time-domain analysis is performed on the BMDRs by applying recorded seismic wave loads. Low reflection boundary conditions are set at the boundaries of the finite element model to simulate the propagation behavior of waves at the

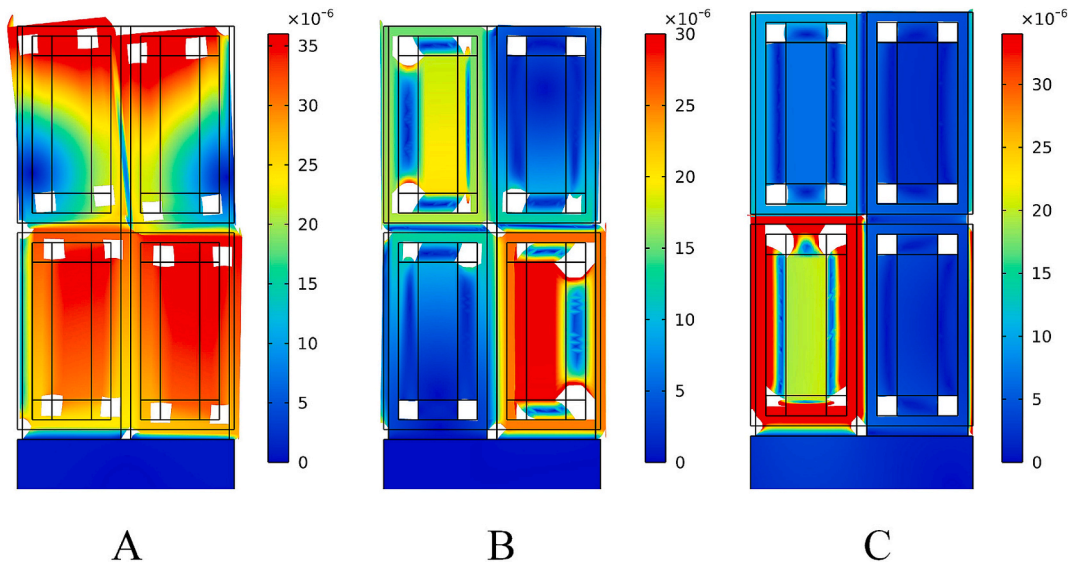
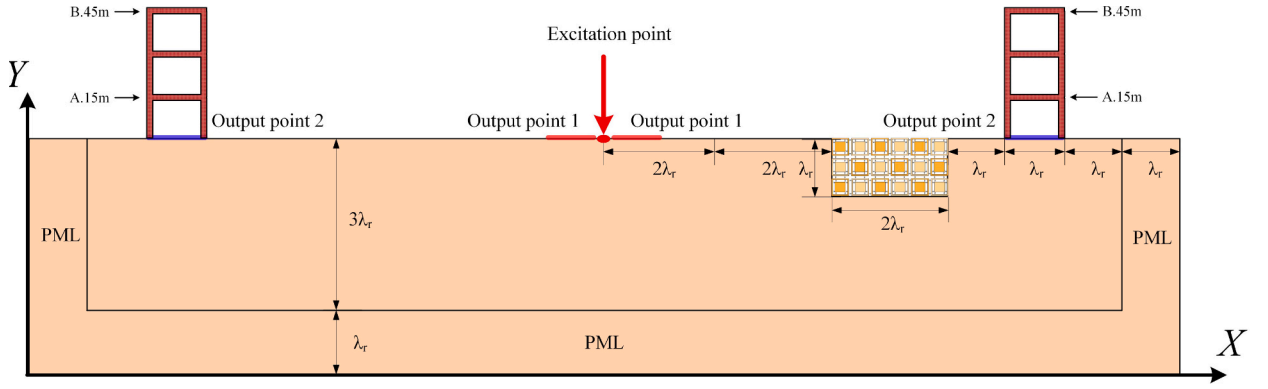
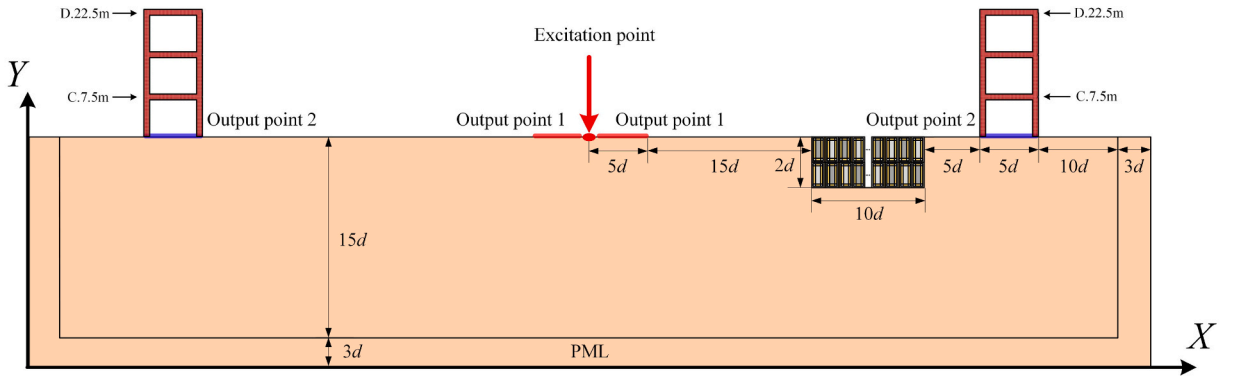


Fig. 7. Modal diagram of BMDR-2.

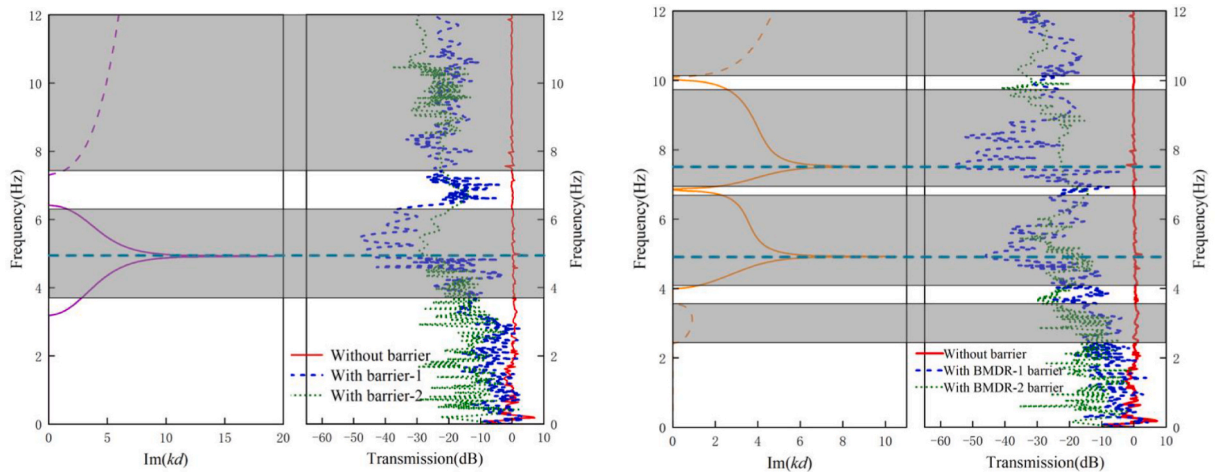


(a)



(b)

Fig. 8. Finite element model of (a) BMDR-1 finite element model; (b) BMDR-2.



(a)

(b)

Fig. 9. Frequency response curves (a) metamaterial with a single resonator; (b) metamaterial with double resonators.

boundaries.

3.2.1. Frequency domain analysis

The frequency response function curve is used to describe the transmission characteristics of the resonant element metamaterial. The transmission spectrum is as follows:

$$\text{FRF} = 20\lg(u_2/u_1) \quad (20)$$

where u_1 represents the average displacement response at output point 1, and u_2 denotes the average displacement response at output point 2.

Furthermore, to ensure that the selected grid size does not affect the numerical simulation results, the current study adopts a triangular grid with a $\text{CR}/f_{\max}/20$ [43] approach, ensuring the numerical solution is convergence. Research has indicated [44] that static loads or displacements applied at excitation points at 0 Hz do not propagate to the response location, the lack of energy transmission at 0 Hz determines the frequency scanning range in this section, which spans from 0.1 Hz to 12 Hz.

A displacement of 0.1 m is applied at the excitation point, and numerical simulations are conducted to obtain the frequency response curves of the metamaterial with single and double resonators. These curves are then compared with the bandgap ranges obtained analytically. Additionally, the frequency response curves without BMDRs are presented, as shown in Fig. 9. It can be observed that the frequency corresponding to the maximum attenuation of the bandgap generated by the analytical solutions of the metamaterials with single or double coincides with the minimum value of the spectral transmission obtained numerically. This frequency can be obtained by substituting the parameter values into Eq. (16). A comparative analysis of single and double-resonator metamaterials reveal that double-resonator configurations not only expand the number of band gaps but also enhance the attenuation effects of elastic waves. When comparing the band gaps generated by local resonance with those arising from Bragg scattering, it is evident that, for elastic waves below 12 Hz, the band gaps induced by local resonance offer superior performance in terms of both width and attenuation effects. By comparing BMDR-1 and BMDR-2, it is evident that BMDR-2, which is closer to the theoretical model, has a better attenuation effect on elastic waves. Moreover, the band gap range generated by localized resonance and Bragg scattering

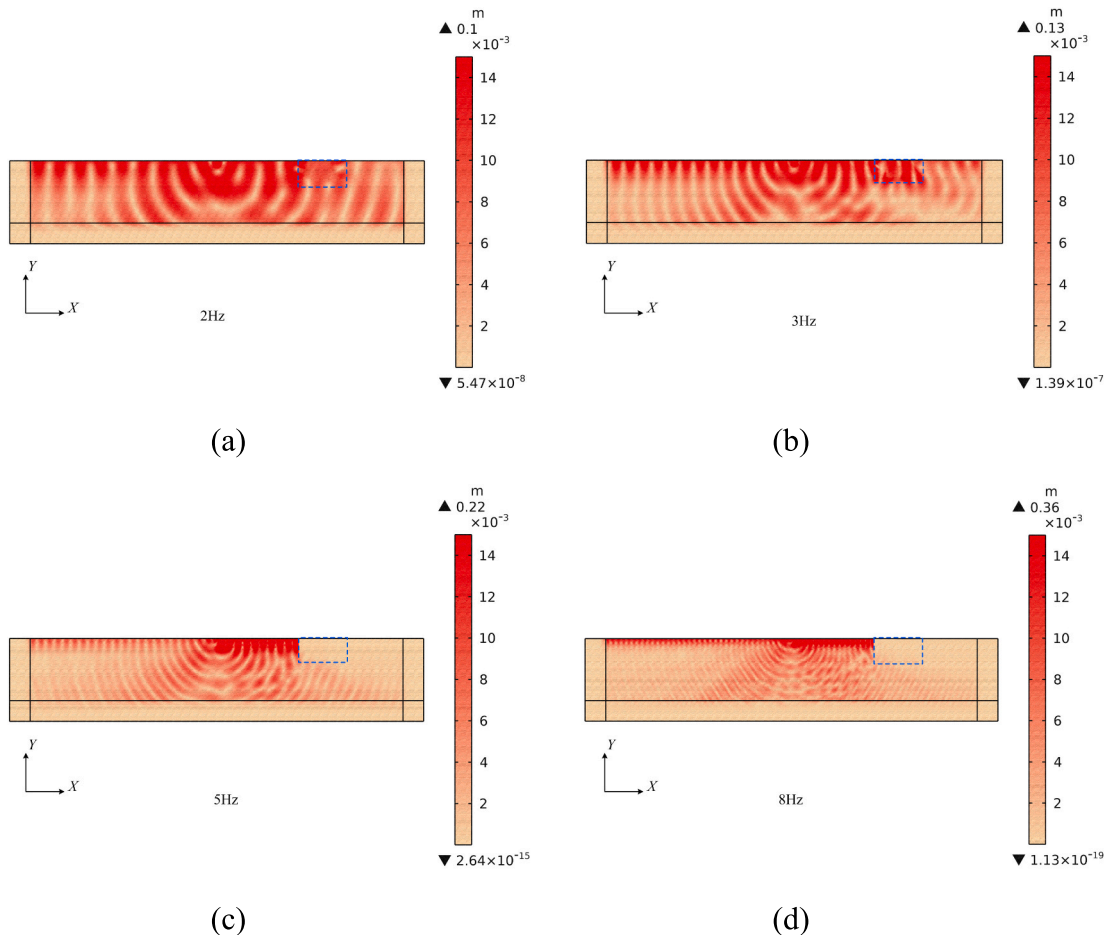


Fig. 10. Displacement field diagram of BMDRs (a) 2 Hz; (b) 3 Hz (c) 5 Hz and (d) 8 Hz.

corresponds well to the frequency range with enhanced attenuation effects in the frequency response curves, thereby confirming the accuracy of the bandgaps in BMDRs, the attenuation effect of the bandgap generated by double resonators and localized resonances is more pronounced.

In terms of attenuation, BMDR-1 exhibits outstanding performance, with an average attenuation of approximately -40 dB within the local resonance bandgap, demonstrating strong signal suppression capabilities. In the Bragg bandgap, the average attenuation of BMDR-1 reaches -25 dB, further confirming its effective attenuation characteristics over a wide frequency range. In contrast, BMDR-2 shows an average attenuation of about -25 dB in the local resonance bandgap, slightly lower than that of BMDR-1, but still demonstrates considerable attenuation. The average attenuation in the Bragg bandgap is -20 dB. Overall, these attenuation results are consistent with theoretical expectations and, compared to the typical attenuation level of around -10 dB in existing studies [45,46], both BMDR-1 and BMDR-2 significantly enhance attenuation performance, reflecting their superiority under multi-band and low-attenuation requirements. This indicates that the proposed design has been validated not only in theoretical models but also shows promising performance in practical applications, with high potential for real-world use.

Fig. 10 shows the displacement field plots of the metamaterial with double resonators under excitations of 2 Hz, 3 Hz, 5 Hz, and 8 Hz frequencies, both outside the bandgap, within the Bragg bandgap, and the localized resonance bandgap. It is evident that when the excitation frequency lies outside the localized resonance bandgap, the attenuation of the excitation is minimal. However, when the excitation frequency falls within the bandgap, a substantial portion of the excitation energy is absorbed, thereby protecting the structures behind the metamaterial. Fig. 11 shows that the energy propagates along the BMDR-1 without notable reduction when the frequency excitation is outside the 2 Hz bandgap. Whereas, when the excitation frequency falls within the 3 Hz Bragg bandgap, only a minimal amount of energy is absorbed. However, when the excitation frequencies are at 5 Hz and 8 Hz, significant resonance occurs in the first two rows of resonators, as most of the energy is absorbed. This effectively protects the subsequent resonators and structures from damage. Similarly, for BMDR-2, when the excitation frequencies are 2 Hz and 3 Hz, the energy propagates along the metamaterial without reduction. Similarly, the first single-cell structure undergoes localized resonance, consuming energy and impeding energy propagation when the excitation frequencies are 5 Hz and 8 Hz, thus forming a super barrier.

3.2.2. Time history analysis

To simulate the seismic isolation and vibration reduction effects of BMDRs during actual seismic events, this section conducted finite element simulations using seismic waves from the PEER Ground Motion Database [47], featuring different dominant frequencies such as the Cape Mendocino, El-Centro, and Anza seismic waves, which are subsequently normalized. The characteristics of these various seismic waves are depicted in Table 3 and Fig. 12.

The dynamic responses of BMDR-1 and BMDR-2 to seismic excitation, with input seismic waves at the excitation point and output seismic waves received at point 2, are illustrated in Table 4 and Table 5, as well as Figs. 13 and 14.

For BMDR-1, when the Cape Mendocino seismic wave is used as the excitation, the attenuation of acceleration response decreased by approximately 23.8 % in the Y-direction at the ground surface, by about 44.8 % in the X-direction, with a decrease of around 38.9 % at point A located at the height of 15 m as shown in Fig. 8(a), and a decrease of roughly 48.2 % at point B situated in the height of 45 m. Moreover, the acceleration response attenuation was approximately 77.2 % in the Y-direction at the ground surface, and around 66.3 % in the X-direction, with a decrease of about 86.2 % at point A and a decrease of around 86.7 % at point B, both at heights of 15 m and 45 m respectively under El-Centro seismic wave with the primary frequency range of 1–4 Hz. The attenuation of the acceleration response was approximately 92.2 % in the Y-direction at the ground surface, and about 92 % in the X-direction at the ground surface, with a decrease of around 97.1 % at point A and a decrease of roughly 96.9 % at point B, both at heights of 15 m and 45 m respectively under Anza seismic wave with the primary frequency range of 5–12 Hz.

Regarding BMDR-2, the acceleration response attenuation was approximately 20.9 % in the Y-direction at the ground surface,

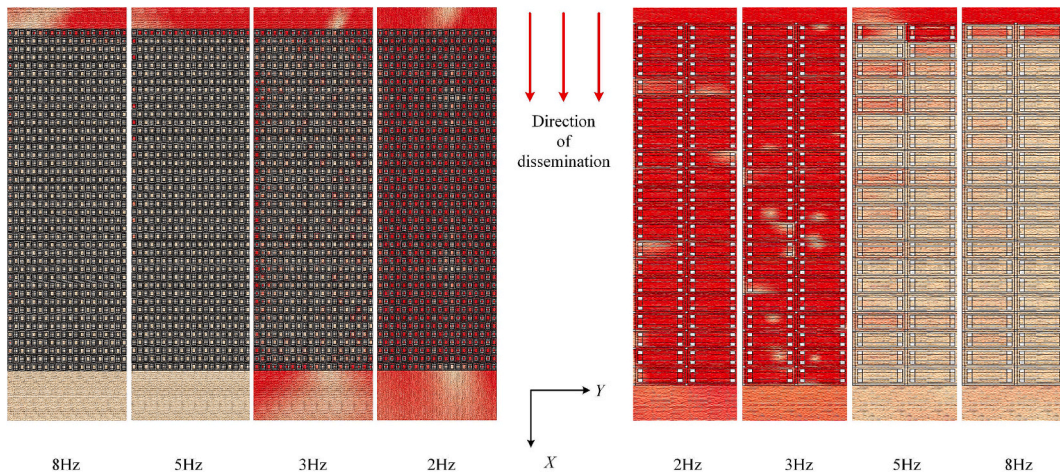


Fig. 11. Local displacement field of BMDRs.

Table 3
Seismic wave characteristics.

Name	Duration / s	Main frequency range / Hz	Amplitude
Cape Mendocino	45	0–2	0.053
El-Centro	50	1–4	0.033
Anza	22	5–12	0.038

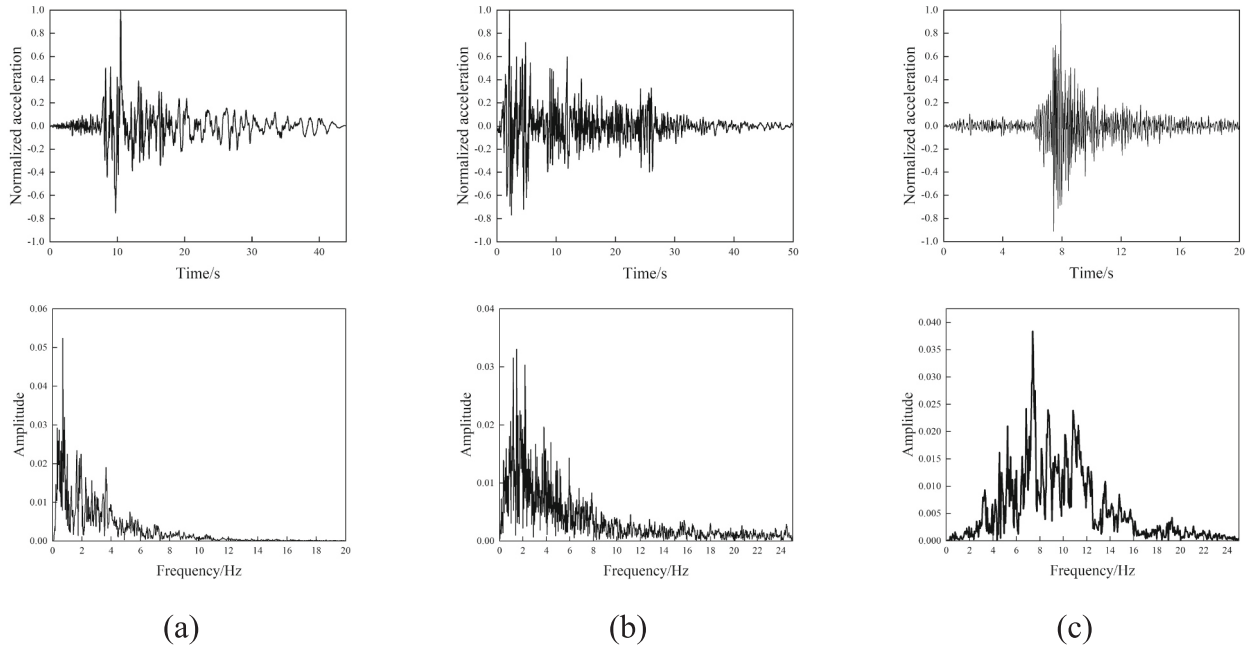


Fig. 12. Seismic wave (a) Cape Mendocino (b)El-Centro seismic wave and (c)Anza.

Table 4
Attenuation reduction of acceleration for BMDR-1.

Name	Surface Y direction	Surface X direction	Point A in the X direction	Point B in the X direction
Cape Mendocino	23.8 %	44.8 %	38.9 %	48.2 %
El-Centro	77.2 %	66.3 %	86.2 %	86.7 %
Anza	92.2 %	92 %	97.1 %	96.9 %

Table 5
Attenuation reduction of acceleration for BMDR-2.

Name	Surface Y direction	Surface X direction	Point C in the X direction	Point D in the X direction
Cape Mendocino	20.9 %	12 %	17.2 %	19.1 %
El-Centro	51 %	30.9 %	26.2 %	40.9 %
Anza	75.1 %	62.4 %	74.9 %	74.8 %

around 12 % in the X-direction, with a decrease of about 17.2 % at point C located at a height of 7.5 m as shown in Fig. 8(b), and a decrease of roughly 19.1 % at point D situated at a height of 22.5 m under Cape Mendocino seismic wave with the primary frequency range of 0–2 Hz. For the primary frequency range of 1–4 Hz for the El-Centro seismic waves, the acceleration response attenuation was approximately 51 % in the Y-direction at the ground surface, around 30.9 % in the X-direction, with a decrease of about 26.2 % at point C and a decrease of roughly 40.9 % at point D, both at heights of 7.5 m and 22.5 m respectively. Furthermore, the acceleration response attenuation was approximately 75.1 % in the Y-direction at the ground surface, and about 62.4 % in the X-direction, with a decrease of around 74.9 % at point C and a decrease of roughly 74.8 % at point D, both at heights of 7.5 m and 22.5 m respectively under Anza seismic wave the primary frequency range of 5–12 Hz.

The results indicate that the attenuation of the acceleration response by BMDR-1 is generally greater than that of BMDR-2, attributed to the arrangement of more metamaterial structures in BMDR-1, which are closer to the theoretical model and provide a

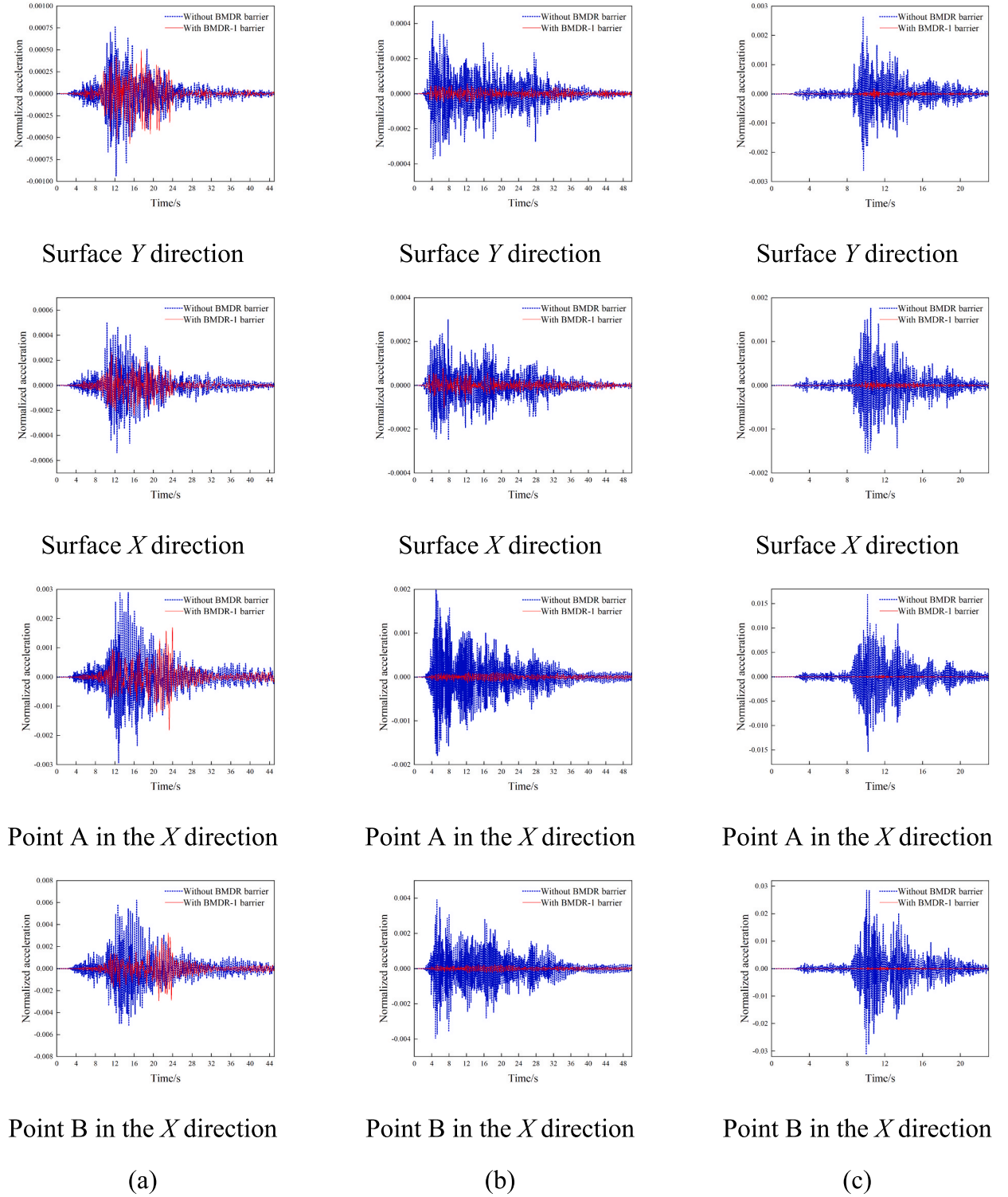
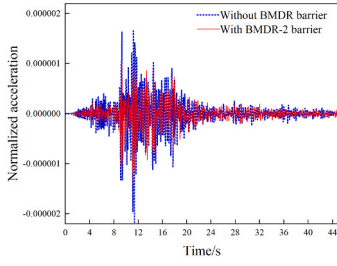
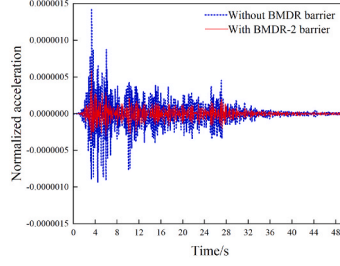
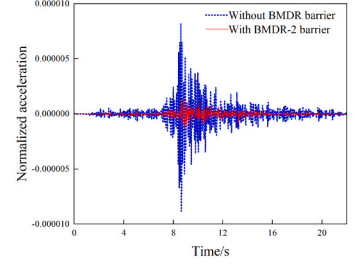
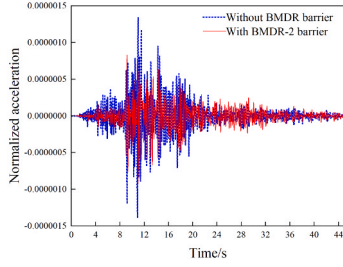
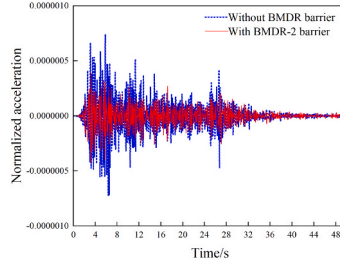
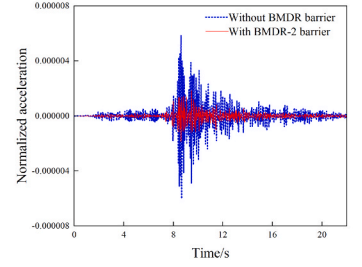
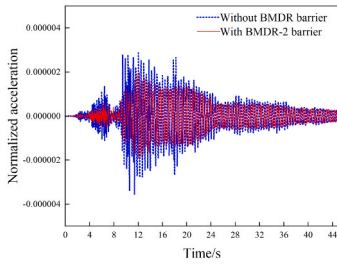
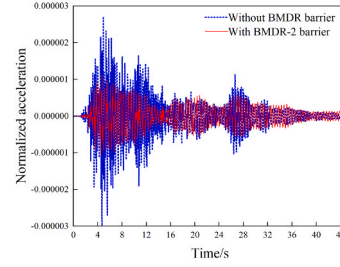
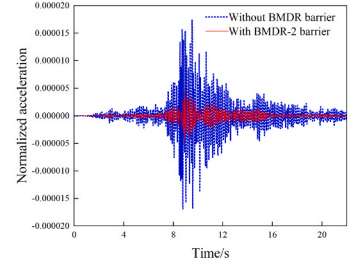
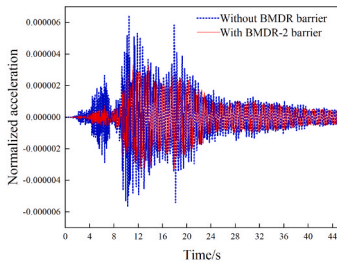
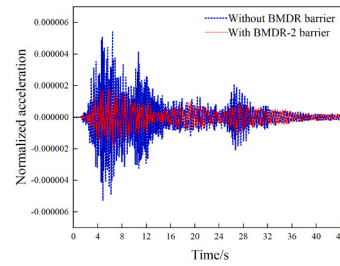
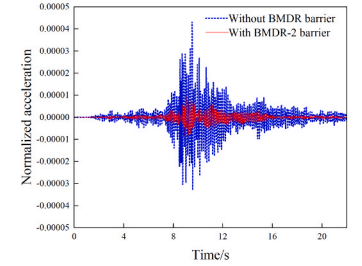


Fig. 13. BMDR-1 acceleration response (a) Cape Mendocino seismic wave (b) El-Centro seismic wave and (c) Anza seismic wave.

more accurate bandgap range, and higher floors exhibit greater acceleration responses. Compared to the typical attenuation levels of around 50 % demonstrated in previous studies[48,49], this study achieves a significant breakthrough in attenuation performance, especially in the dynamic response analysis of building floors, where it showcases innovative potential. However, the increase in amplitude after attenuation by the metamaterial barrier against incoming seismic waves is not significant. For seismic waves with

Surface Y directionSurface Y directionSurface Y directionSurface X directionSurface X directionSurface X directionPoint C in the X directionPoint C in the X directionPoint C in the X directionPoint D in the X directionPoint D in the X directionPoint D in the X direction

(a)

(b)

(c)

Fig. 14. BMDR-2 acceleration response (a) Cape Mendocino seismic wave; (b) El-Centro seismic wave; (c) Anza seismic wave.

lower frequency ranges such as Cape Mendocino, Eq. (19) suggests that enhancing the mass of steel blocks and reducing the elasticity modulus of rubber can lower the frequency range of the bandgap, thus improving the attenuation effectiveness of the metamaterial barrier against lower frequency seismic waves.

4. Conclusion

In this paper, based on the results of theoretical derivation and analysis, two kinds of buried metamaterials with double resonators are proposed. Firstly, the dispersion relations of the metamaterial structures are obtained through theoretical derivation and numerical simulation. Secondly, frequency and time domain analysis are used to attenuation effects of the metamaterial barriers. Finally, the attenuation effects are verified by using seismic waves excitation. The conclusions are obtained as follows:

(1) The dispersion curves of both BMDR-1 and BMDR-2 are in high agreement with theoretical solutions, confirming the precision of the theoretical models used. BMDR-1 features three well-defined band gaps at 2.06–3.17 Hz, 4.04–6.43 Hz, and 6.89–9.60 Hz, while BMDR-2 generates band gaps at 2.07–3.47 Hz, 4.03–6.30 Hz, and 7.17–9.50 Hz. These band gaps are critically important as they represent frequency ranges where wave propagation is effectively blocked. The first band gap in both materials originates from Bragg scattering, while the second and third band gaps are attributed to local resonance phenomena.

(2) The numerical attenuation domains of BMDR-1 and BMDR-2 align closely with the theoretically determined band gap ranges, which serves to further validate the accuracy and reliability of the calculated band gaps. Additionally, the displacement field diagrams reveal that when the excitation frequency falls within the band gap range, the metamaterial layers experience intense local resonance with the incoming seismic waves, preventing their transmission to subsequent layers. This mechanism ensures that the structure remains protected from the damaging effects of seismic waves, highlighting the superior protective capabilities of the metamaterials in seismic applications.

(3) BMDRs achieve an average attenuation of approximately more than 70 % for seismic waves at the dominant frequency within the local resonance band gap and more than 30 % within the Bragg band gap. Moreover, it is crucial to note that neither of these metamaterials exhibits any amplification effect for seismic waves outside the band gap, ensuring that they do not inadvertently intensify unwanted frequencies. This characteristic greatly enhances their practicality and efficiency, making them highly reliable for seismic protection applications.

CRedit authorship contribution statement

Chunfeng Zhao: Writing – original draft, Investigation, Conceptualization. **Zhiwei Gao:** Conceptualization. **Fan Kong:** Writing – review & editing, Validation, Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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